Thermistor

Aim

To understand the working principle behind the thermistor.

Analysis

The Steinhart-Hart Equation is used to convert resistance reading of a thermistor to thermodynamic temperature.

$$\frac{1}{T} = A + B \ln R + C(\ln R)^3$$

The formula was derived using fundamental principles of solid state physics. NTCs are polycrystalline semiconductor devices; the resistance should follow the classical temperature activated form.

$$R = R_0 e^{B\left(\frac{1}{T} - \frac{1}{T_0}\right)}$$

which on solving gives

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B}\left(\ln \frac{R}{R_0}\right)$$

which dissolves into a linear relation

$$\frac{1}{T} = a_0 + a_1 \ln R$$

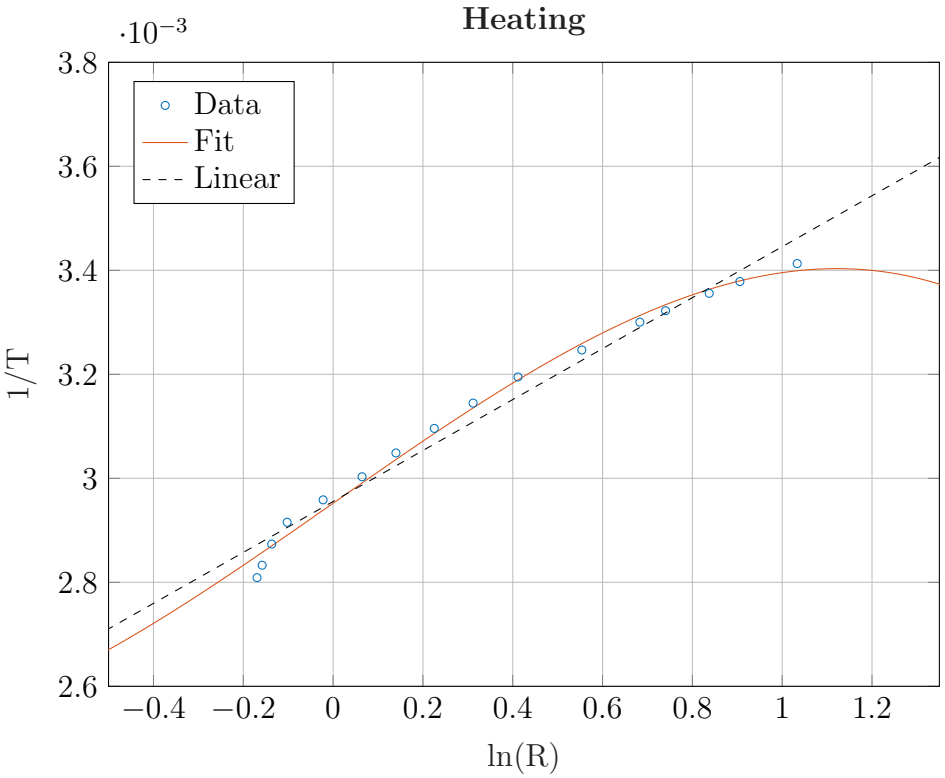
This is a good enough relation for small changes in resistance. For larger deviations, B itself becomes temperature dependent and we apply series expansion to get more accurate results.

$$\frac{1}{T} = \sum_{i=0}^{n} a_i \ln^i R$$

We find that a_2 is statistically insignificant and there is no loss in precision by terminating the expansion after $n = 3$. This gives us back the Steinhart-Hart Equation. We use both linear and cubic fits to approximate the relationship between $\ln R$ and $\frac{1}{T}$.

Safety precautions : The induction stove must not go beyond 800 watts. The pipes carring steam must not be bent. Use gloves to touch hot equipment.

The data is different for heating and cooling the materials and follows a hysteresis. Hence we perform the regression separately.

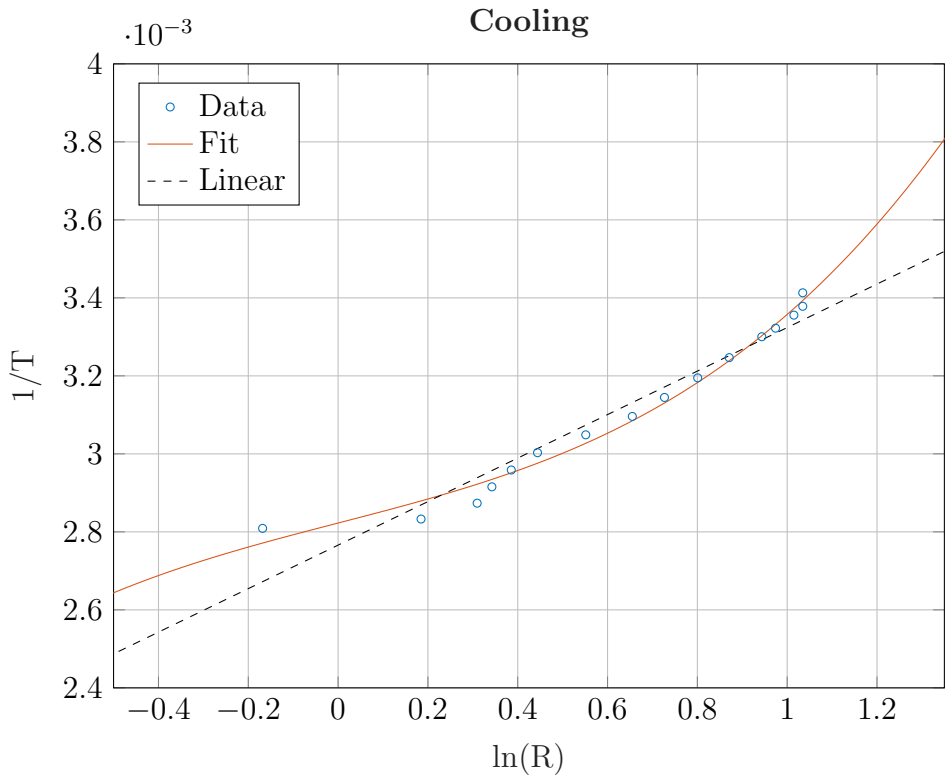


For the heating curve we get the following regression parameters
For cubic fit:

$$A = 2.95208 \times 10^{-3}$$
$$B = 0.60321 \times 10^{-3}$$
$$C = -0.159792 \times 10^{-3}$$
$$R^2 = 0.9927$$

For linear fit:

$$a_0 = 2.95547 \times 10^{-3}$$
$$a_1 = 0.489919 \times 10^{-3}$$
$$R^2 = 0.9769$$



For the cooling curve we get the following regression parameters
For cubic fit:

$$A = 2.82265 \times 10^{-3}$$
$$B = 0.298851 \times 10^{-3}$$
$$C = 0.236043 \times 10^{-3}$$
$$R^2 = 0.9856$$

For linear fit:

$$a_0 = 2.76602 \times 10^{-3}$$
$$a_1 = 0.557824 \times 10^{-3}$$
$$R^2 = 0.9422$$

Result

We have verified that the resistance of the thermistor depends on the temperature wrt the Steinhart-Hart Equation. The thermistor is linear for small temperature ranges.